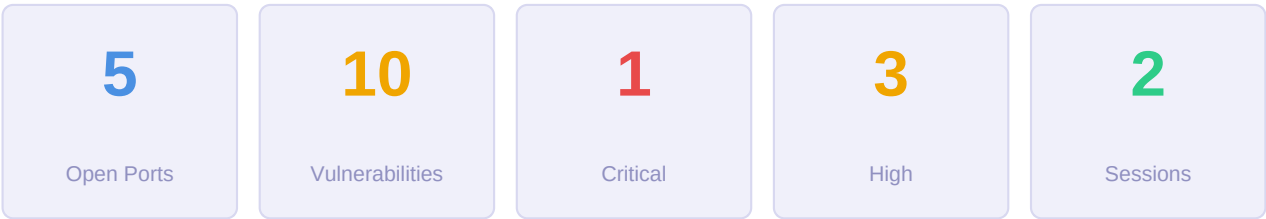


PENTEST REPORT

Reconnaissance and Vulnerability Analysis

Target: 10.129.227.181

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1. Executive Summary

Attack Surface Summary

The attack surface is **CRITICAL** due to the presence of the MS17-010 (EternalBlue) vulnerability (CVE-2017-0143), which allows remote code execution on the target system. The system is running an outdated and unsupported operating system (Windows XP), and SMBv1 is likely in use, which is known to be highly vulnerable. Additionally, guest authentication is enabled and message signing is disabled, further increasing the risk of exploitation.

Detected Services and Versions

- **Port 135:** Microsoft Windows RPC
- **Port 137:** netbios-ns
- **Port 139:** Microsoft Windows netbios-ssn
- **Port 445:** Microsoft Windows XP microsoft-ds (SMBv1)

No service versions were explicitly identified beyond the OS (Windows XP).

Key Findings

- **MS17-010 (EternalBlue) vulnerability** (CVE-2017-0143) is present, allowing remote code execution.
 - **SMBv1** is likely in use, which is known to be vulnerable to attacks like EternalBlue.
 - **SMB message signing is disabled**, making the system vulnerable to relay attacks.
 - **Guest authentication is enabled**, allowing unauthenticated access to shared resources.
 - **Outdated OS:** Windows XP is no longer supported and has known security risks.
 - **Inconsistent system time** may indicate misconfiguration or compromise.
-

Recommended Next Steps

1. **Exploit MS17-010 (CVE-2017-0143)** using the Metasploit module `exploit/windows/smb/ms17_010_eternalblue`.
 - Command: `use exploit/windows/smb/ms17_010_eternalblue`
 - Set RHOSTS to `10.129.227.181`
 - Set PAYLOAD to `windows/meterpreter/reverse_tcp` (if meterpreter is desired) or another appropriate payload.
2. **Verify SMBv1 usage** by checking the SMB protocol version during exploitation or using tools like `nmap` with the `--script smb-protocols` option.

3. Disable guest authentication and enable SMB message signing to mitigate relay attacks and unauthorized access.

4. Upgrade the operating system immediately, as Windows XP is no longer supported and poses a significant security risk.

5. Isolate the system from the network until it can be patched or replaced, as it is highly vulnerable to exploitation.

2. Confirmed Attack Vectors

4 known vulnerabilities identified by automatic correlation of detected service versions against the exploit knowledge base. These vectors are **confirmed** by exact service version and do not depend on LLM analysis.

[CRITICAL] EternalBlue SMBv1 Remote Code Execution Vulnerability

CVE-2017-0143

Port: 445 · **Service:** microsoft-ds

[HIGH] SMBv1 in Use with Guest Authentication Enabled

Port: 445 · **Service:** microsoft-ds

[HIGH] NTP Integer Overflow Vulnerability

CVE-2009-0159

Port: 123 · **Service:** ntp

[MEDIUM] NTP Buffer Overflow Vulnerability

CVE-2009-1252

Port: 123 · **Service:** ntp

3. Ports and Services Detected

5 open port(s) detected on target.

Port	Service	Version	Proto	Notes
123	ntp	—	TCP	
135	Microsoft	Windows RPC	TCP	
137	netbios-ns	—	TCP	
139	Microsoft	Windows netbios-ssn	TCP	
445	Microsoft	Windows XP microsoft-ds	TCP	■ Sensitive service

4. CVE Vulnerabilities — Full Detail

6 CVE(s) identified across exposed services (5 from NVD · 0 from knowledge base).

Severity	Count	Typical Impact
HIGH	1	Privilege escalation, unauthorised data access
MEDIUM	5	Denial of service, partial information disclosure

Port 445 — Microsoft Windows XP microsoft-ds

[HIGH] CVE-2017-0143

8.8

The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windows 7 SP1; Windows 8.1; Windows Server 2012 Gold and R2; Windows RT 8.1; and Windows 10 Gold, 1511, and 1607; and Windows Server 2016 allows remote attackers to execute arbitrary code via crafted packets, aka "Windows SMB Remote Code Execution Vulnerability." This vulnerability is different from those described in CVE-2017-0144, CVE-2017-0145, CVE-2017-0146, and CVE-2017-0148.

Published	2017-03-17
CVSS Vector	CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:H
Attack Vector	NETWORK
CWE	NVD-CWE-noinfo

Exploitation context:

Remotely exploitable — no physical access or local presence required. RCE confirmed: search Exploit-DB or GitHub for PoC. Metasploit: exploit/windows/smb/ms17_010_eternalblue Exploit-DB Difficulty: easy. EternalBlue: RCE in SMBv1 (Windows).

References:

packetstormsecurity.com
packetstormsecurity.com
securityfocus.com
NVD

Port 123 — ntp

[MEDIUM] CVE-2009-0159**6.8**

Stack-based buffer overflow in the cookedprint function in ntpq/ntp.c in ntpq in NTP before 4.2.4p7-RC2 allows remote NTP servers to execute arbitrary code via a crafted response.

Published	2009-04-14
CVSS Vector	AV:N/AC:M/Au:N/C:P/I:P/A:P
Attack Vector	NETWORK
CWE	CWE-119

Exploitation context:

Remotely exploitable — no physical access or local presence required. Buffer overflow: may allow arbitrary code execution.

References:

ftp.netbsd.org
bugs.pardus.org.tr
lists.apple.com
NVD

[MEDIUM] CVE-2009-1252**6.8**

Stack-based buffer overflow in the crypto_recv function in ntp_crypto.c in ntpd in NTP before 4.2.4p7 and 4.2.5 before 4.2.5p74, when OpenSSL and autokey are enabled, allows remote attackers to execute arbitrary code via a crafted packet containing an extension field.

Published	2009-05-19
CVSS Vector	AV:N/AC:M/Au:N/C:P/I:P/A:P
Attack Vector	NETWORK
CWE	CWE-119

Exploitation context:

Remotely exploitable — no physical access or local presence required. Buffer overflow: may allow arbitrary code execution.

References:

ftp.netbsd.org
lists.opensuse.org
Red Hat
NVD

[MEDIUM] CVE-2009-3563**6.4**

ntp_request.c in ntpd in NTP before 4.2.4p8, and 4.2.5, allows remote attackers to cause a denial of service (CPU and bandwidth consumption) by using MODE_PRIVATE to send a spoofed (1) request or (2) response packet that triggers a continuous exchange of MODE_PRIVATE error responses between two NTP daemons.

Published	2009-12-09
CVSS Vector	AV:N/AC:L/Au:N/C:N/I:P/A:P
Attack Vector	NETWORK
CWE	NVD-CWE-Other

Exploitation context:

Remotely exploitable — no physical access or local presence required. DoS: may disrupt the service.

References:

ftp.netbsd.org
aix.software.ibm.com
Debian Security
NVD

[MEDIUM] CVE-2004-0657**5.0**

Integer overflow in the NTP daemon (NTPd) before 4.0 causes the NTP server to return the wrong date/time offset when a client requests a date/time that is more than 34 years away from the server's time.

Published	2004-08-06
CVSS Vector	AV:N/AC:L/Au:N/C:N/I:P/A:N
Attack Vector	NETWORK
CWE	CWE-190

Exploitation context:

Remotely exploitable — no physical access or local presence required.

References:

marc.info
kb.cert.org
exchange.xforce.ibmcloud.com
NVD

[MEDIUM] CVE-2009-0021**5.0**

NTP 4.2.4 before 4.2.4p5 and 4.2.5 before 4.2.5p150 does not properly check the return value from the OpenSSL EVP_VerifyFinal function, which allows remote attackers to bypass validation of the certificate chain via a malformed SSL/TLS signature for DSA and ECDSA keys, a similar vulnerability to CVE-2008-5077.

Published	2009-01-07
CVSS Vector	AV:N/AC:L/Au:N/C:P/I:N/A:N
Attack Vector	NETWORK
CWE	CWE-287

Exploitation context:

Remotely exploitable — no physical access or local presence required. Authentication failure: access possible without valid credentials.

References:

lists.apple.com

lists.opensuse.org

lists.opensuse.org

NVD

5. Metasploit Exploitation Results

10 module(s) executed · 4 vulnerable · 2 session(s) opened · 0 error(s)

Execution Plan

#	CVE	CVSS	Module	Result
1	CVE-2008-0015	9.0	exploit/windows/browser/msvidctl_mpeg2	NO RESULT
2	CVE-2008-0015	9.0	exploit/windows/browser/msvidctl_mpeg2	NO RESULT
3	CVE-2008-0015	9.0	exploit/windows/browser/msvidctl_mpeg2	NO RESULT
4	CVE-2017-0143	8.8	exploit/windows/smb/ms17_010_psexec	SESSION
5		7.5	exploit/windows/smb/ms17_010_psexec	SESSION

SESSION OPENED

6. Execution Timing

Total scan duration: **14m 43s** (883.9s)

Phase	Duration	% of Total
Model warm-up	56s	6.4%
Phase 0 — TCP Discovery	42s	4.9%
Phase 1 — Version Detection	46s	5.2%
Phase 2 — Adaptive NSE	1m 44s	11.9%
Phase 3 — Firewall Detection	5s	0.6%
Phase 4 — UDP Scan	29s	3.3%
Phase 5 — Web Fuzzing	0s	0.0%
Phase 6 — Metasploit	8m 00s	54.4%
Phase 7 — Searchsploit + Creds	15s	1.8%
Total	14m 43s	100%

7. Execution Log

8 step(s) executed during reconnaissance.

✓ Step 1 — Full TCP SYN scan (all 65535 ports)

```
nmap -Pn -sS -p- --min-rate 1000 --max-retries 2 10.129.227.181
```

→ **Found:** 135 (msrpc), 139 (netbios-ssn), 445 (microsoft-ds)

→ SMB service detected (port 445) which may indicate a Windows system or Samba server.

✓ Step 2 — Version + OS detection on 135,139,445

```
nmap -Pn -sV -O --version-intensity 5 -sC --script-timeout 60s -p 135,139,445 -oX /home/marina/Desktop/pentestM/RAG/outputs/nmap_sv_10_129_227_181.xml 10.129.227.181
```

→ **Found:** 135 (Microsoft Windows RPC), 139 (Microsoft Windows netbios-ssn), 445 (Windows XP microsoft-ds)

→ The host is running an outdated Windows XP operating system, which is no longer supported and poses significant security risks.

→ SMBv1 is likely in use (as SMB2 protocol negotiation failed), which is known to be vulnerable to attacks such as EternalBlue.

→ Message signing is disabled on the SMB service, which can allow for man-in-the-middle attacks.

→ The system time appears to be inconsistent, which may indicate a misconfigured or compromised system.

→ SMB guest authentication enabled

✓ Step 3 — Check for MS17-010 (EternalBlue) vulnerability on port 445

```
nmap -Pn -sV --script smb-vuln-ms17-010 -p 445 10.129.227.181 --script-timeout 30s
```

→ **Found:** 445 (Microsoft Windows XP microsoft-ds)

→ The target is vulnerable to the MS17-010 (EternalBlue) exploit (CVE-2017-0143).

→ This is a critical remote code execution vulnerability in Microsoft SMBv1 servers.

CVEs detected by nmap:

CVE-2017-0143 · score: None · port: 445

✓ Step 4 — ACK scan — firewall detection

```
nmap -Pn -sA -p 135,139,445 10.129.227.181
```

✓ Step 5 — UDP scan — top 100 ports by frequency

```
nmap -Pn -sU --top-ports 100 --open --max-retries 1 10.129.227.181
```

→ **Found:** 123 (ntp), 137 (netbios-ns)

✓ **Step 6 — UDP scan — exploit-focused service ports**

```
nmap -Pn -sU -p 53,67,69,111,123,137,138,161,162,500,623,1434,1900,4500,5353,11211,17185  
--open --max-retries 1 10.129.227.181
```

→ Found: 123 (ntp), 137 (netbios-ns)

✓ **Step 7 — MSF exploitation phase**

```
msfconsole (multiple modules)
```

✗ **Step 7 — Searchsploit + credential phase**

```
searchsploit / paramiko / ftplib
```